

Human Violence And Object Detection

Automated Test Execution Report

1. Report Information

Project: Human Violence And Object Detection

Date: 24-April-2026 06:54 PM

Framework: Python unittest + unittest.mock

Total Modules: 10

Total Test Cases: 94

Result: 94 Passed | 0 Failed | 0 Skipped | 100.0% Pass Rate

Execution Time: 30.15 seconds

2. Module: Login

Test ID	Description	Status	Time
TC-L01	Admin login with valid credentials	PASS	44ms
TC-L02	Admin login with wrong password	PASS	0ms
TC-L03	Login with empty username should be rejected	PASS	0ms
TC-L04	Login with empty password should be rejected	PASS	0ms
TC-L05	Login with both fields empty should be rejected	PASS	0ms
TC-L06	Operator login with valid credentials	PASS	0ms
TC-L07	Login with non-existent username should fail	PASS	0ms
TC-L08	Login with SQL injection attempt should fail safely	PASS	1ms
TC-L09	Press Enter key in password field should trigger login	PASS	0ms
TC-L10	Closing LoginWindow should exit the application	PASS	0ms

Login: 10/10 passed (100%)

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3. Module: Registration

Test ID	Description	Status	Time
TC-R01	Register operator with all valid fields	PASS	4ms
TC-R02	Register with duplicate username should fail	PASS	0ms
TC-R03	Register with username 'admin' should fail	PASS	0ms
TC-R04	Register with empty Full Name should fail	PASS	0ms
TC-R05	Register with less than 10 digit mobile should fail	PASS	0ms
TC-R06	Register with letters in mobile should fail	PASS	0ms
TC-R07	Register with password shorter than 6 characters should fail	PASS	0ms
TC-R08	Register with mismatched passwords should fail	PASS	0ms
TC-R09	Register with invalid email format should fail	PASS	0ms
TC-R10	Cancel registration should not save data	PASS	0ms
TC-R11	Login should succeed immediately after successful registration	PASS	0ms

Registration: 11/11 passed (100%)

4. Module: Database

Test ID	Description	Status	Time
TC-D01	Database file should be created on initialization	PASS	272ms
TC-D02	All 6 tables should exist after initialization	PASS	4ms
TC-D03	Default admin user should be seeded on first init	PASS	4ms
TC-D04	Admin password should be stored as bcrypt hash, not plain text	PASS	4ms
TC-D05	log_detection should insert a row into detections table	PASS	12ms
TC-D06	log_alert should insert a row linked to a detection	PASS	398ms
TC-D07	log_snapshot should insert a row into snapshots table	PASS	8ms
TC-D08	log_recording should insert a row into recordings table	PASS	8ms
TC-D09	fetch_detections should respect the limit parameter	PASS	52ms

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TC-D10	Two threads writing simultaneously should not corrupt DB	PASS	13ms
TC-D11	get_operator_email should return correct email after registration	PASS	416ms

Database: 11/11 passed (100%)

5. Module: Camera

Test ID	Description	Status	Time
TC-C01	Camera should open successfully with index 0	PASS	1ms
TC-C02	cap.read() should return ret=True and a valid numpy frame	PASS	1ms
TC-C03	Frame should have 3 colour channels (BGR)	PASS	1ms
TC-C04	Frame should not be all zeros (all black)	PASS	3ms
TC-C05	Camera read should complete in < 500ms	PASS	2ms
TC-C06	30 consecutive frames should all read successfully	PASS	2ms
TC-C07	After release, camera should not return valid frames	PASS	1ms
TC-C08	Opening non-existent camera index 99 should fail gracefully	PASS	1ms
TC-C09	Resizing frame to (640, 480) should produce correct shape	PASS	25ms

Camera: 9/9 passed (100%)

6. Module: Knife Detection

Test ID	Description	Status	Time
TC-K01	Knife model file best.pt should exist	PASS	0ms
TC-K02	YOLO('best.pt') should load without errors	PASS	6704ms
TC-K03	model.predict() should return results list without crashing	PASS	4ms
TC-K04	Prediction on knife image should return at least 1 bounding box	PASS	3ms
TC-K05	Confidence score should be between 0.0 and 1.0	PASS	2ms
TC-K06	Bounding box coordinates should satisfy $x1 < x2$, $y1 < y2$, all ≥ 0	PASS	3ms
TC-K07	model.names should return a dict with class names	PASS	0ms
TC-K08	Knife inference should complete in < 500ms	PASS	2ms

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TC-K09	Model should return empty detections on a blank/black frame	PASS	1ms
TC-K10	Model should not crash on a very small frame (10x10)	PASS	0ms

Knife Detection: 10/10 passed (100%)

7. Module: Violence Detection

Test ID	Description	Status	Time
TC-V01	Violence model file best2.pt should exist	PASS	0ms
TC-V02	YOLO('best2.pt') should load without errors	PASS	0ms
TC-V03	model.predict() should return results list without crashing	PASS	5ms
TC-V04	Prediction on violent content should return at least 1 detection	PASS	4ms
TC-V05	Confidence score should be between 0.0 and 1.0	PASS	2ms
TC-V06	Violence inference should complete in < 500ms	PASS	3ms
TC-V07	Model should return empty detections on a blank frame	PASS	2ms
TC-V08	Both YOLO models should load and predict on the same device	PASS	4ms

Violence Detection: 8/8 passed (100%)

8. Module: Email Service

Test ID	Description	Status	Time
TC-E01	Email config should have sender, host, and port set correctly	PASS	0ms
TC-E02	When EMAIL_ENABLED=False, no email should be sent	PASS	14ms
TC-E03	MIME message should have correct From, To, Subject fields	PASS	0ms
TC-E04	Email should include attached snapshot images	PASS	2ms
TC-E05	Email should include attached video clip	PASS	1ms
TC-E06	Attaching non-existent file should not crash	PASS	0ms
TC-E07	Only first 4 snapshots should be attached even if 6 are provided	PASS	100ms
TC-E08	send_direct_alert should return a Thread and not block	PASS	4ms
TC-E09	Email should be sent to both operator and admin emails	PASS	7ms

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TC-E10	Empty recipient list should fall back to config.EMAIL_RECEIVER	PASS	3ms
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Email Service: 10/10 passed (100%)

9. Module: Recording Service

Test ID	Description	Status	Time
TC-RC01	recordings/ directory should be created if it does not exist	PASS	0ms
TC-RC02	start_recording should create an MP4 clip file path	PASS	3004ms
TC-RC03	Video clip filename should contain a timestamp	PASS	3004ms
TC-RC04	Recorder should write frames_needed = CLIP_DURATION_SEC * VIDEO_FPS	PASS	4009ms
TC-RC05	Second start_recording call while recording should be ignored	PASS	3232ms
TC-RC06	Queue maxsize=200 should silently drop excess frames	PASS	1ms
TC-RC07	Recording should run in a daemon background thread	PASS	3004ms
TC-RC08	is_recording should reflect recording state correctly	PASS	5115ms

Recording Service: 8/8 passed (100%)

10. Module: GUI / Dashboard

Test ID	Description	Status	Time
TC-G01	LoginWindow should be created on application startup	PASS	3ms
TC-G02	AdminDashboard should open after successful login	PASS	9ms
TC-G03	Sidebar should contain all 10 navigation items	PASS	44ms
TC-G04	Click 'Start Detection' should create IntegrationWindow	PASS	2ms
TC-G05	Click 'Stop Detection' should set _running = False	PASS	2ms
TC-G06	Logout should destroy dashboard and show LoginWindow	PASS	2ms
TC-G07	Exit should destroy root window, terminating the application	PASS	2ms
TC-G08	View Alerts should query and display detections from DB	PASS	2ms
TC-G09	View Recordings should show saved video clips	PASS	3ms
TC-G10	View Snapshots should list snapshot images	PASS	1ms

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TC-G11	Speed Profile should return profiler data with avg/min/max/calls	PASS	13ms
TC-G12	Theme colours should be valid hex codes	PASS	0ms

GUI / Dashboard: 12/12 passed (100%)

11. Module: Speed Profiler

Test ID	Description	Status	Time
TC-P01	Profiler should record approximately correct timing	PASS	103ms
TC-P02	After 3 runs, min <= avg <= max should hold	PASS	92ms
TC-P03	reset() should clear all recorded data	PASS	11ms
TC-P04	5 threads measuring simultaneously should all be recorded	PASS	25ms
TC-P05	summary_lines() should return formatted strings	PASS	23ms

Speed Profiler: 5/5 passed (100%)

12. Summary

Module	Total	Pass	Fail	Skip	Rate
Login	10	10	0	0	100.0%
Registration	11	11	0	0	100.0%
Database	11	11	0	0	100.0%
Camera	9	9	0	0	100.0%
Knife Detection	10	10	0	0	100.0%
Violence Detection	8	8	0	0	100.0%
Email Service	10	10	0	0	100.0%
Recording Service	8	8	0	0	100.0%
GUI / Dashboard	12	12	0	0	100.0%
Speed Profiler	5	5	0	0	100.0%
OVERALL	94	94	0	0	100.0%

Total execution time: 30.15 seconds

13. Conclusion

All automated unit tests for the Human Violence And Object Detection system have passed successfully. A total of 94 tests were executed across 10 modules using Python's unittest framework, with zero failures and zero skipped tests, yielding a 100% pass rate.

The test suite validates critical system functionality including admin and operator authentication, secure bcrypt password storage, SQL injection prevention, SQLite database integrity with thread-safe concurrent writes, OpenCV camera capture and frame processing, YOLO-based knife and violence detection with bounding box validation, SMTP email alerting with MIME attachments, background video clip recording, full Tkinter GUI navigation, and millisecond-precision performance profiling.



ALL TESTS PASSED
SYSTEM STATUS: HEALTHY

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